

Question

1. What is a conditional statement in programming?

A conditional statement in programming is used to make decisions based on certain conditions. It allows a program to execute different blocks of code depending on whether a condition is true or false.

Conditional statements help control the flow of execution and make programs more intelligent and interactive.

Question

- 2. What does an if statement do?

An if statement checks whether a specified condition is true. If the condition evaluates to true, the code inside the if block is executed. If the condition is false, the program skips that block and continues with the next statement.

Question

- 3. What is the purpose of an else block?

The else block provides an alternative set of instructions to execute when the condition in the if statement is false. It ensures that some action is taken when the primary condition is not satisfied.

Question

- 4. When is an else block executed?

An else block is executed only when the condition associated with the if statement evaluates to false. If the if condition is true, the else block is ignored.

Question

- 5. What is a Boolean expression?

A Boolean expression is an expression that evaluates to either true or false. It is commonly used in conditional statements to make decisions. Examples include comparisons such as `a > b` or `x == y`.

Question

- 6. Write the syntax of a simple if statement.

The syntax of a simple if statement is:

C++

```
if(condition)
```

```
{
```

```
    // statements
```

```
}
```

The statements inside the braces execute only when the condition is true.

Question

- 7. What operator is commonly used to test equality?

The equality operator `==` is commonly used to test whether two values are equal. It returns true if both values are the same and false otherwise.

Question

- 8. What does the modulus (%) operator return?

The modulus (%) operator returns the remainder obtained after dividing one number by another. For example, $10 \% 3$ returns 1 because 1 is the remainder.

Question

- 9. How can you check if a number is even?

A number can be checked for evenness by using the modulus operator. If `number % 2 == 0`, the number is even because it leaves no remainder when divided by 2.

Question

- 10. How can you check if a number is odd?

A number can be checked for oddness by using the modulus operator. If $\text{number} \% 2 \neq 0$, the number is odd because it leaves a remainder when divided by 2.

Question

- 11. What is the purpose of else-if?

The purpose of an else-if statement is to test multiple conditions one after another. If the first if condition is false, the program checks the else-if condition. This allows the program to choose from several possible actions based on different conditions.

Question

- 12. What is a nested if statement?

A nested if statement is an if statement placed inside another if or else block. It is used when a second condition needs to be checked only after the first condition has been satisfied. Nested if statements help in making more detailed decisions.

Question

- 13. Can an if statement exist without an else?

Yes, an if statement can exist without an else statement. In such cases, the code inside the if block executes only when the condition is true. If the condition is false, the program simply skips that block and continues executing the remaining statements.

Question

- 14. What is the output when a false condition is tested in an if statement without else?

When a false condition is tested in an if statement without an else block, no output or action occurs from that if statement. The program ignores the statements inside the if block and moves on to the next instruction.

Question

- 15. What is the role of braces `{}` in C/C++ conditionals?

In C/C++ conditionals, braces `{}` are used to group multiple statements into a single block. They indicate which statements belong to the `if`, `else`, or `else-if` block. Braces improve readability and help prevent logical errors.

Question

- 16. What does `>` mean?

The operator `>` means "greater than." It compares two values and returns true if the value on the left is larger than the value on the right. Otherwise, it returns false. For example, `10 > 5` evaluates to true.

Question

- 17. What does `<` mean?

The operator `<` means "less than." It checks whether the value on the left is smaller than the value on the right. If the condition is true, it returns true; otherwise, it returns false. For example, `3 < 7` is true.

Question

- 18. What does `>=` mean?

The operator `>=` means "greater than or equal to." It returns true if the left value is either greater than or exactly equal to the right value. For example, `18 >= 18` and `20 >= 18` are both true.

Question

- 19. What does `<=` mean?

The operator `<=` means "less than or equal to." It checks whether the left value is smaller than or equal to the right value. If either condition is satisfied, it returns true. For example, `5 <= 5` and `4 <= 5` are true.

Question

- 20. What does != mean?

The operator != means "not equal to." It compares two values and returns true if they are different. If both values are the same, it returns false. For example, 10 != 20 is true, while 10 != 10 is false.

Question

- 21. What is a switch statement?

A switch statement is a decision-making structure used to select one block of code from many alternatives. It compares the value of an expression with different case labels and executes the matching case. It is often used instead of long if-else-if ladders when dealing with multiple constant values.

Question

- 22. What is a case label in a switch statement?

A case label in a switch statement represents a specific value that is compared with the switch expression. If the expression matches a case label, the statements under that case are executed.

Question

- 23. Why is break used in switch?

The break statement is used in a switch statement to terminate the execution of the current case. It prevents the program from continuing into the next cases and helps maintain correct program flow.

Question

- 24. What happens if break is omitted?

If the break statement is omitted, the program continues executing the statements in the following cases until it encounters a break or reaches the end of the switch block. This behavior is called fall-through.

Question

- 25. What is the default case in switch?

The default case in a switch statement is executed when none of the case labels match the value of the switch expression. It acts as the final alternative, similar to the else block in an if-else statement.

Question

- 26. Can switch compare strings in C++?
No, a switch statement cannot directly compare strings in traditional C++. It is mainly used with integral types such as integers and characters. String comparisons are usually performed using if-else statements.

Question

- 27. Which statement is better for many constant choices: if-else or switch?

For many constant choices, a switch statement is generally better than an if-else ladder because it is easier to read, organize, and maintain. It also improves code clarity when there are many possible fixed values.

Question

- 28. How do you test whether a value is positive?

A value can be tested for positivity by checking whether it is greater than zero. The condition used is:

C++

`value > 0`

If this condition is true, the value is positive.

Question

- 29. How do you test whether a value is negative?

A value can be tested for negativity by checking whether it is less than zero. The condition used is:

C++

`value < 0`

If the condition is true, the value is negative.

Question

- 30. How do you test whether a value is zero?

A value can be tested for zero by checking whether it is equal to zero. The condition used is:

C++

`value == 0`

If this condition evaluates to true, the value is exactly zero.

Question

- 31. Write a condition to check if age is at least 18.

To check if a person is at least 18 years old, the condition used is:

C++

`age >= 18`

If this condition is true, the person is considered eligible as an adult. This type of condition is commonly used in programs related to voting, driving licenses, and registrations.

Question

- 32. Write a condition to check if marks are above 50.

To check if marks are above 50, the following condition is used:

C++

marks > 50

If the condition evaluates to true, it means the student has scored more than 50 marks. Such conditions are often used to determine pass status or eligibility criteria.

Question

- 33. What is decision making in programming?

Decision making in programming is the process of choosing different actions based on specific conditions. It enables a program to analyze situations and execute appropriate instructions. Conditional statements such as if, else-if, else, and switch are used for decision making.

Question

- 34. Can nested if statements have multiple levels?

Yes, nested if statements can have multiple levels. One if statement can be placed inside another if statement repeatedly as needed. This allows programmers to handle complex decision-making situations involving several conditions.

Question

- 35. What is logical AND (&&)?

Logical AND (&&) is an operator used to combine two or more conditions. It returns true only when all the conditions involved are true. If any one condition is false, the entire expression becomes false.

Question

- 36. What is logical OR (||)?

Logical OR (||) is an operator that combines multiple conditions. It returns true if at least one of the conditions is true. It returns false only when all the conditions are false.

Question

- 37. What is logical NOT (!)?

Logical NOT (!) is an operator that reverses the result of a condition. It changes true to false and false to true. It is commonly used when the opposite of a condition is required.

Question

- 38. How do you check if a year is divisible by 4?

To check whether a year is divisible by 4, the following condition is used:

C++

```
year % 4 == 0
```

If this condition is true, the year is divisible by 4 because the remainder after division is zero.

Question

- 39. What is a leap year?

A leap year is a year that contains 366 days instead of 365 days. In the Gregorian calendar, February has 29 days in a leap year. Leap years occur to keep the calendar synchronized with the Earth's revolution around the Sun.

Question

- 40. What is the first condition for a leap year?

The first condition for a leap year is that the year must be divisible by 4. If a year is not divisible by 4, it cannot be a leap year.

Question

- 41. How do you find the largest of two numbers?

To find the largest of two numbers, compare them using the greater-than operator:

C++

```
if(a > b)
    largest = a;
else
    largest = b;
```

The number with the greater value is considered the largest. This method is commonly used in programming to compare numerical values.

Question

- 42. How do you find the largest of three numbers?

To find the largest of three numbers, compare all three numbers using if-else statements. The number that is greater than the other two is considered the largest. This method helps determine the maximum value among three given numbers.

Question

- 43. What is an if-else-if ladder?

An if-else-if ladder is a series of conditional statements used to test multiple conditions one after another. The program executes the block corresponding to the first condition that evaluates to true. If none of the conditions are true, the else block is executed.

Question

- 44. What is the difference between if and switch?

The main difference between if and switch is that an if statement can evaluate complex conditions and ranges using logical and relational operators, whereas a switch statement compares a single expression against fixed constant values. Switch statements are generally more organized for handling many constant choices.

Question

- 45. Can switch use ranges directly?

No, a switch statement cannot use ranges directly in C/C++. Each case label must contain a single constant value. To test ranges such as marks between 80 and 90, if-else statements are usually preferred.

Question

- 46. What is fall-through in switch?

Fall-through in a switch statement occurs when the break statement is omitted. As a result, after executing one case, the program continues executing the statements in the following cases until it encounters a break or reaches the end of the switch block.

Question

- 47. What data type is usually used for conditions?

The Boolean data type (bool) is usually used for conditions in programming. It stores only two possible values: true and false. Conditional statements use these values to determine which block of code should be executed.

Question

- 48. Why are conditional statements important?

Conditional statements are important because they allow programs to make decisions and respond differently to different situations. They improve the flexibility and usefulness of programs by enabling them to choose appropriate actions based on given conditions.

Question

- 49. Give one real-life example of a conditional statement.

A real-life example of a conditional statement is: If it is raining, carry an umbrella; otherwise, do not carry one. This demonstrates how decisions are made based on

Question

- 50. Name four conditional structures covered.

The four conditional structures covered are if statement, if-else statement, else-if ladder, and switch statement. These structures are used in programming to make decisions based on different conditions. They help control the flow of a program by executing specific blocks of code according to whether conditions are true or false. They are essential for creating logical and interactive programs.